

MOBILE CHARGER WITH MULTIPLE VOLTAGE REGULATION & PROTECTION TECHINQUES.

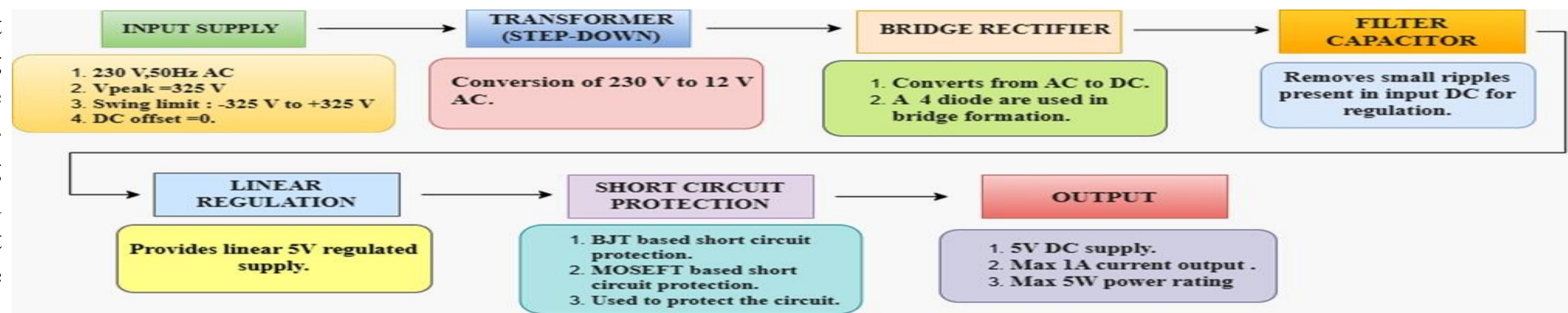
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Problem statement

To design, simulate, and implement an AC–DC mobile charging circuit that converts a 230 V, 50 Hz supply into a stable 5 V DC output using discrete electronic components. The system uses a transformer, bridge rectifier, filter circuit, and Zener-based voltage regulation to obtain low-ripple DC output. The project incorporates multiple current handling circuits and BJT- and MOSFET-based short-circuit protection mechanisms to ensure safe and reliable operation. The complete circuit is implemented on PCB and tested to verify stable output and effective protection.

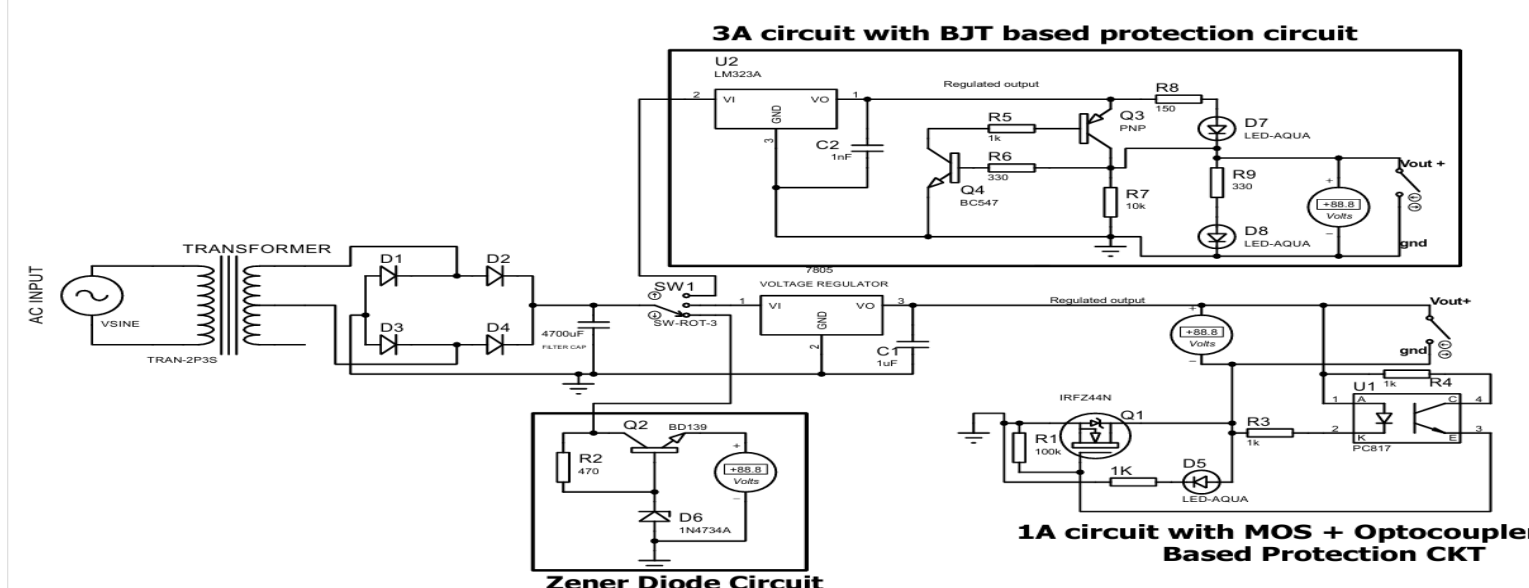
Electronic System Design (ESD): Detailed Block Diagram



System Design

- The system uses a step-down transformer to reduce high-voltage AC to a safe level.
- A bridge rectifier converts AC into DC, and a filter capacitor reduces ripple.
- A 7805 / LM323A / Zener diode as a linear voltage regulator maintains a constant 5 V DC output.
- A short-circuit protection circuit using BJT/MOSFET and optocoupler continuously monitors output current and protects the circuit during fault conditions.

Simulation



Results and Discussions

- The designed linear regulator–based mobile charger successfully converts 230 V AC input into a regulated 5 V DC output suitable for mobile charging applications.
- The rectifier and filter circuit effectively reduce ripple, and the voltage regulator maintains a stable output under normal load conditions.
- During a short-circuit or overload condition, the electronic protection circuit activates immediately, disconnecting the load and preventing damage to the components.
- Experimental results confirm safe operation, reliable voltage regulation, and effective short-circuit protection

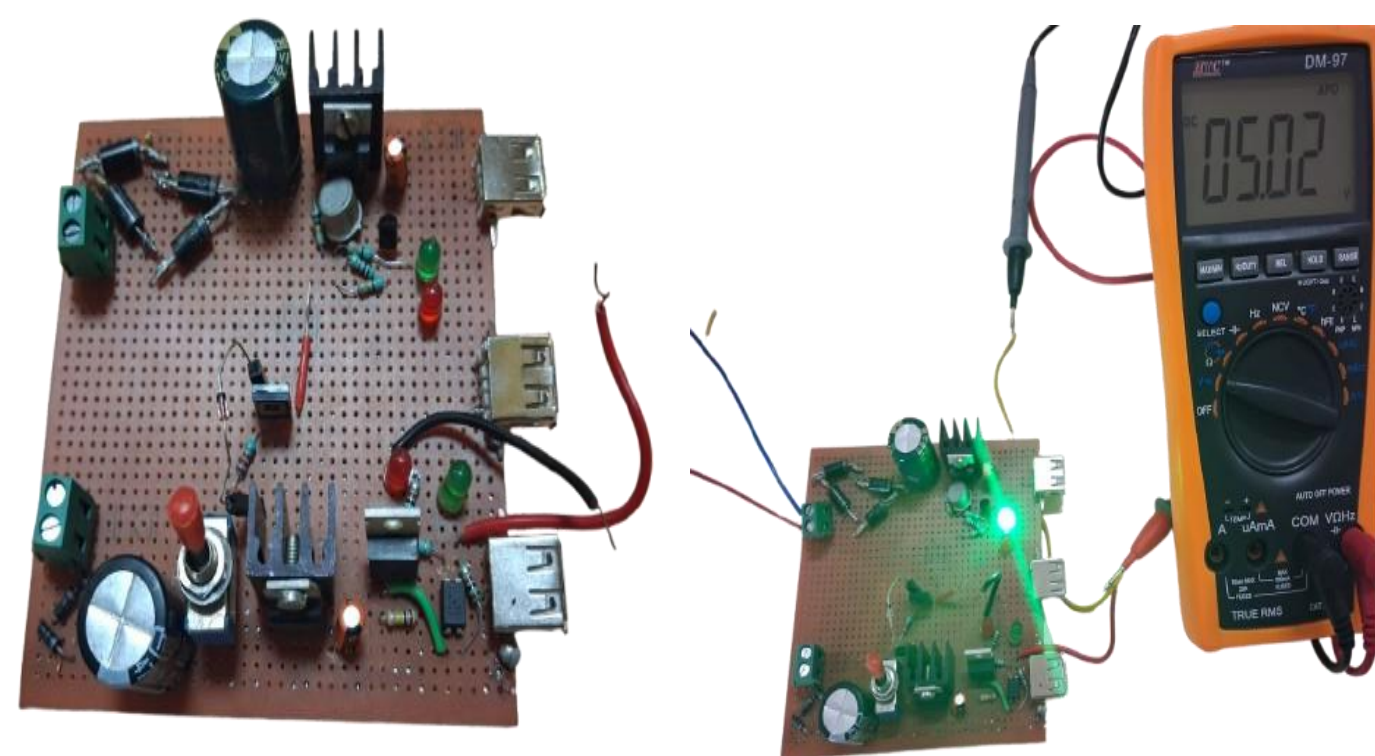
Alternate solutions

- ❖ SMPS-Based Mobile Charger :Compact and efficient but complex and costly.
- ❖ Buck Converter (DC–DC Converter)-Based Mobile Charger :Offers high efficiency and reduced power loss, but requires complex control circuitry and careful design.
- ❖ **Linear Regulator–Based Mobile Charger with Protection** :Selected for its stable output , protection, simplicity, and suitability for low-power mobile charging applications.

Specifications and Design

- Input Voltage: Single Phase 230V, 50Hz AC input
- Output Voltage: 5V $\pm$ 5% Regulated DC power supply.
- Maximum Output Current: 1 A -3A.
- Output voltage ripple: < 50mV.
- Transformer Specifications :12 V – 0 V – 12 V center tapped transformer.
- Regulator Specifications :5V output supply, 1A-3A maximum current.

Hardware realization



Conclusion and Reflections

- ✓ The linear regulator–based mobile charger was successfully designed and implemented to demonstrated stable voltage regulation, low ripple, and effective electronic short-circuit protection, ensuring safe and reliable operation.
- ✓ This project enhanced practical understanding of power supply design, rectification, voltage regulation, and protection techniques, and provided valuable hands-on experience in circuit

References

- Ghodke, Manisha, et al. "Design and development of stabilized regulated fixed and variable DC power supply unit (PSU) with short circuit protection." *Indian Journal of Scientific Research* (2015): 267-273.
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